PRAMAAN प्रमाण

Proof, not a verdict.

Face liveness and presentation-attack detection for the MOSIP Registration Client, desktop and Android, that leaves evidence a server can re-check.

MOSIP Decode 2026 · Problem Statement 04

Team: Bhavya Garg · github.com/GargBhavya-tech/pramaan · MPL-2.0

Evidence-first prototype

The gap nobody names

The preview is unsigned. The capture is signed. Nothing links them.



- **Capture swap:** whoever controls the preview can pass liveness with one face and enrol another.
- **Android:** the client never sees preview frames at all; the device app shows the stream and returns only the signed capture.
- **Injected video** (OBS, deepfake pipelines) is outside ISO/IEC 30107. **Insiders** can log in with a supervisor's photo.
- **No audit trail:** a client-side pass/fail cannot be re-checked by the server later or re-scored with a better model.

Our answer: two interfaces, seven questions

The problem statement asks to define the frame-streaming interface and the liveness/PAD interface. We did, and bound them together.

FaceFrameSource
frames *and* the signed capture from the same object, so the capture can be bound to what was watched

LivenessEngine
frames in; states, guidance and a decision out; never a score on the screen; fails closed

1	2	3	4	5	6	7
Real sensor? source integrity	One usable face quality gate	Looks alive? passive, asks nothing	Attack marks? print, screen, flat	Fusion accept / challenge / reject	Live answer? random challenge	Same face? binding + signed evidence

- **Mode C (desktop):** the engine runs in the client, on the unsigned M-JPEG stream (3 fps guaranteed, so every cue is gated on the measured frame rate).
- **Mode D (Android):** the engine ships as a library inside the SBI device app, where the frames actually are.

What is different

Each idea is built and tested. None is claimed beyond what the claim sheet allows.

Capture binding

Four locks before any accept: nonce echoed in the device-signed capture, frame hash chain, identity continuity, time. Works on **unmodified SBI**.

Evidence, not a verdict

A signed JWS in the face BIR. One command re-verifies it; a newer model can re-score stored crops. Merkle transparency log.

Cues that fail differently

Passive, moiré, flash reflection (20.7-bit random colour code; author-tested on 1 setup), scene, stereo. Each abstains when the station lacks hardware.

Injection defence

Device identity chain, OS camera registrations (OBS found via DirectShow), JPEG stream fingerprint. Not a renamed-driver defence.

Inclusion by design

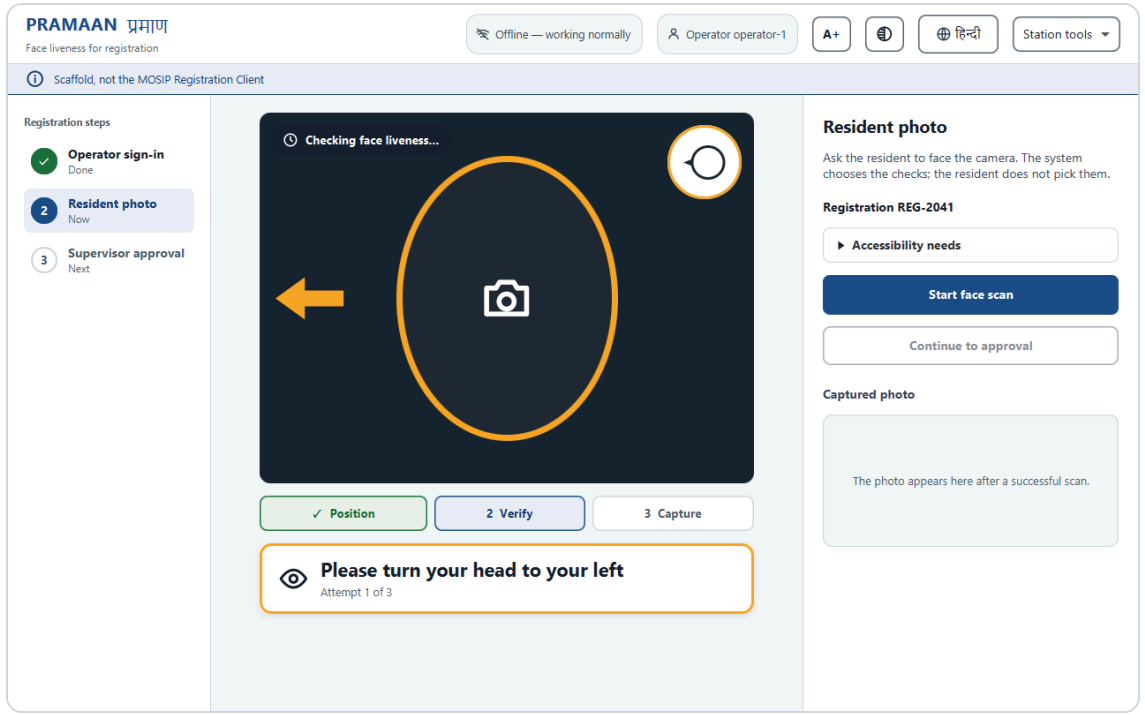
Blink, pulse and pupil reflex only ever vote *for* life. Hindi, icons, speech, accessibility profiles, infant mode, dual-control exceptions.

An SBI proposal

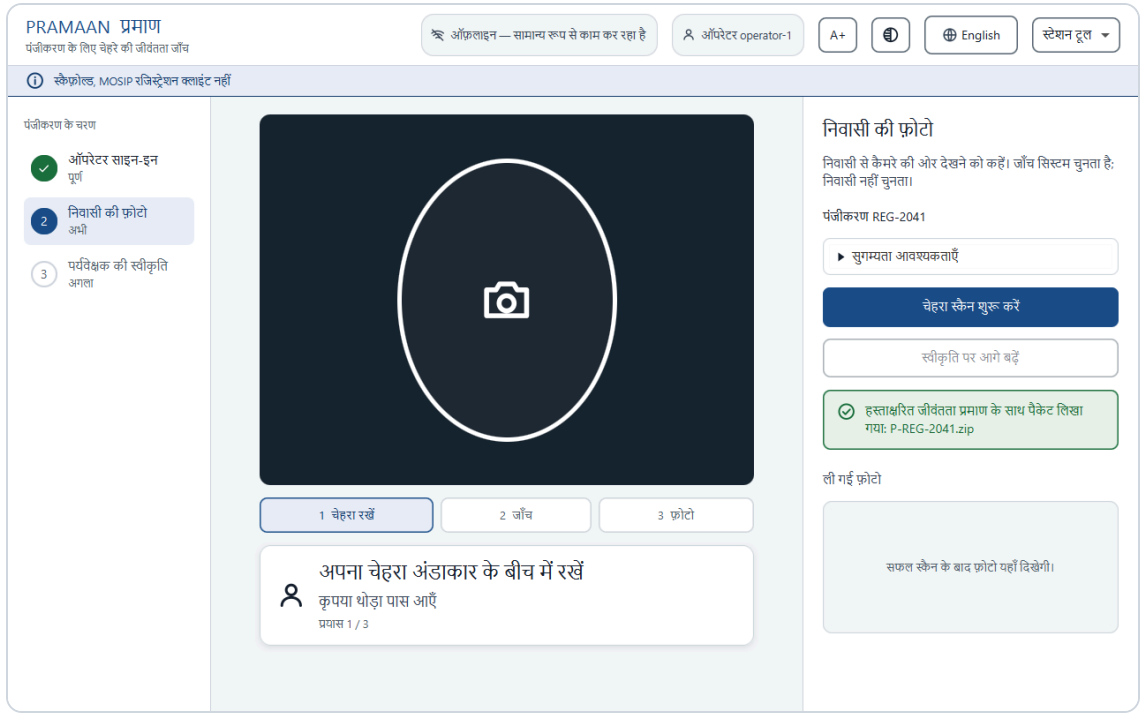
An optional Liveness Profile (signed preview, signed light log, signed burst) with a runnable reference device. Our own draft; not yet posted to MOSIP.

The station, as built

Rendered from the real code, offscreen. One plain sentence, an icon, no score. No live camera in these images.



Resident capture. Step rail, one instruction, Offline chip, the scaffold labelled as a scaffold.



Hindi, larger text, high contrast are one click. (Full Devanagari translation, verified by a native Hindi speaker.)

Fails closed. If the policy, the operating-point table or a model does not verify, nothing can be registered and the screen says what to check (*docs/media/fails-closed.png*). Failures show the resident one generic sentence and never the cause.

Evidence and limits

Measured, tested and disclosed: what we can show, and what we cannot yet.

Tested and measured

- **738 automated tests pass** (0 failures, strict CI on Linux & Windows)
- **Hardware measured:** Models and camera timed on one phone; OBS veto shown on a Windows webcam list
- **Flash challenge:** 1,679,616 sequences = **20.7 bits** (≤ 3 changes/sec, WCAG 2.3.1 compliant)
- **Capture binding:** 4 cryptographic locks; capture-swap attack rejected in every test
- **Insider threat analytics:** 6 in 40 at 2% baseline yields z ≈ 5.87; operator self-enrollment blocked
- **Audit trail:** Merkle transparency log matching Certificate Transparency test vectors

Scientific Discipline & Limits

- **Scaffold, not full Client:** Evaluation scaffold; no live session on certified SBI device yet
- **Android replay limit:** Passive model fooled by screen replay (0.98 score) & flash needs ≥ 3 fps (phone got 2.7–2.9 fps); replay defence on Android rests on challenge alone
- **Inclusive UX:** Native Devanagari Hindi verified; accessibility profiles for elderly & disabled residents
- **Red team findings:** automated fuzzer found moiré blind spots; fusion voting mitigates them on synthetic replays. The moiré cue measured on one phone camera does not separate live from replay (AUC 0.67), so it abstains.

Self-Red-Teaming. On synthetic screen replays with the passive model assumed fooled and no flash light, the fusion accepted **4 of 85** display pitches; with the flash cue, **0**. Logged as R-03 with R-01 (moiré blind pitches); the fix (an accept needs a flash or stereo vote, else a challenge) is enforced for operators and supervisors.

Impact and next steps

A hook-based design for today's SBI ecosystem. Evidence so far: synthetic replays and one-phone timings; field calibration is next.

For MOSIP

- Closes the capture-swap gap on unmodified SBI; leaves signed proof a server can re-verify
- Patches that apply cleanly (checked in CI) to MOSIP registration-client v1.2.0.2 and android-registration-client at 8f4643a. Desktop adds a challenge-prompt screen; Android is hooks only. Plus a standalone evaluation scaffold.
- MPL-2.0 licence; JavaFX 11 matching MOSIP Registration Client runtime
- Asymmetric physiological cues (positive-only) prevent accidental exclusion of darker skin or elderly residents

MOSIP Standards Proposal

- SBI Liveness Profile: backward-compatible extension for signed streams and light logs

Next, in order

1. **Field calibration:** Execute the consented multi-subject protocol via built-in conformal calibrator for final site-specific operating points
2. **SBI device vendor testing:** Partner with certified biometric device manufacturers to test hardware profile integration
3. **Registration Client pilot:** Run end-to-end user trials with field registration operators

Repo: github.com/GargBhavya-tech/pramaan
Every claim and its command: docs/claim-sheet.md